University of Moratuwa
Department of Computer Science and Engineering

AutoVaultLK

User Manual

Guests and Buyers

Second-Hand Vehicle Marketplace with Intelligent Search and Automated Inventory Processing

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1. Introduction

1.1 Purpose and Scope

This manual describes the use of AutoVaultLK, a web-based marketplace for second-hand vehicles in Sri Lanka, by guests and buyers. It explains how vehicles are found, compared and saved, and how a buyer account is created and maintained. The functions available to dealers and administrators are described in separate manuals, and installation and technical details are outside the scope of this document.

1.2 System Overview and User Categories

AutoVaultLK brings together listings published by dealers and presents them through a search facility that accepts ordinary sentences, tolerates spelling errors and ranks the results by relevance. The system was developed as Project 11 by Group 25 of the Department of Computer Science and Engineering, University of Moratuwa. Its functional requirements are recorded in the Software Requirements Specification [2], and its design is described in the Software Architecture Document [3].

Two categories of user are covered by this manual. A guest is any person who uses the public pages without an account, and a buyer is a guest who has created a free account. The functions available to each category are summarised in Table 1, and the usual sequence of activities is shown in Fig. 1. A visit normally begins on the home page, continues through searching and the inspection of a vehicle, and ends with a call to the dealer. Signing in is required only when a vehicle is to be saved for later.

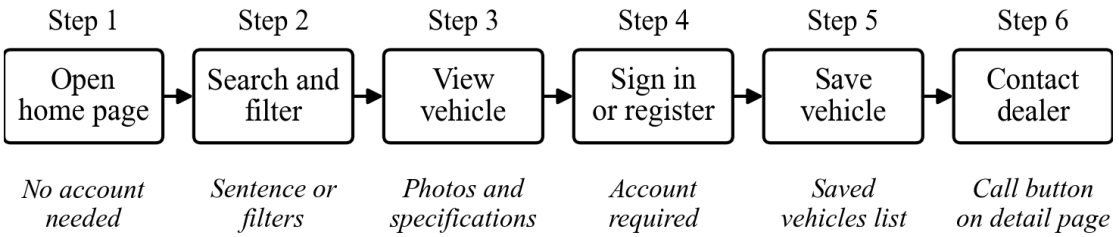


Fig. 1. Typical journey of a guest or buyer.

Table 1. Functions available to guests and buyers.

Function	Guest	Buyer
Browse, search and filter vehicles	Yes	Yes
View vehicle details and contact the dealer	Yes	Yes
Save vehicles to a personal list	No	Yes
Reset a forgotten password	Not applicable	Yes

1.3 Conventions

The names of buttons, links, tabs and fields are printed in bold exactly as they appear on screen. All names, vehicles and figures shown in the screen captures are sample data and will differ from those seen in live use.

2. Getting Started

2.1 Requirements and Access

The platform is used through a current version of a standard web browser, such as Chrome, Edge, Firefox or Safari, on a computer, tablet or telephone, and requires an internet connection. A valid email address is needed only when an account is created. The public pages open without signing in, so that a guest can begin searching immediately after opening the web address of the platform.

2.2 Home Page and Navigation

The home page, part of which is shown in Fig. 2, presents a search panel with a free-text box and make and budget selectors, live counts of listed vehicles, verified dealers and makes, and a strip of the most recent listings that can be filtered by vehicle category. Further down, category tiles allow a type of vehicle to be browsed in one step. Every public page carries the same header, which contains the logo, the main menu (**Home**, **Browse**, **Just listed** and **Dealers**), a search icon that expands into a search box, a heart icon that opens the saved list, and a **Sign in** link. The **Add listing** button leads dealers to their own portal and is not used by buyers.

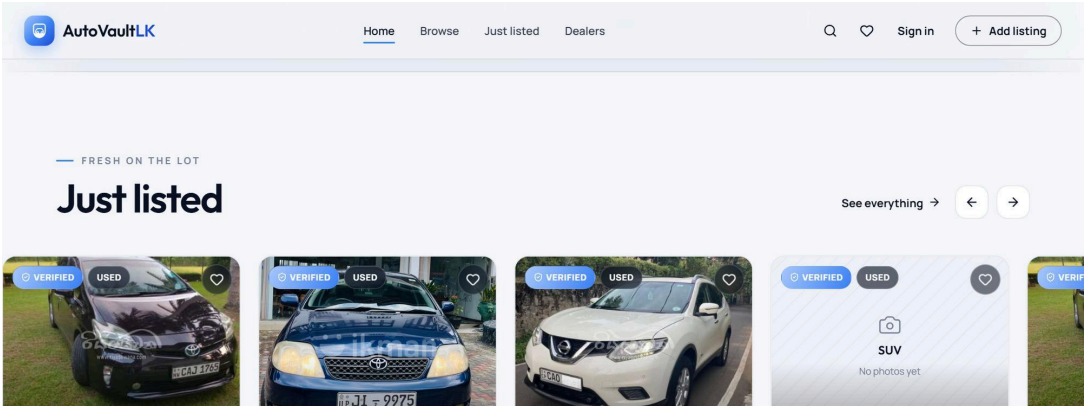


Fig. 2. Home page with the header and the Just listed strip.

After sign-in, the **Sign in** link is replaced by a round account button that shows the first letter of the buyer's name. Selecting it opens the menu shown in Fig. 3, which displays the name and email address of the account and offers **Saved vehicles** and **Sign out**.

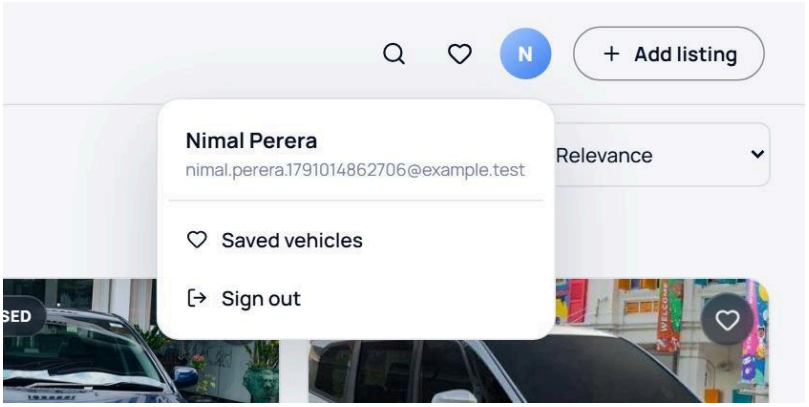


Fig. 3. Account menu opened from the account button.

2.3 Use on Mobile Devices

The interface adapts to small screens, as illustrated in Fig. 4. The main menu collapses behind a menu button, results are displayed in a single column, and the filter panel slides over the page when **Filters** is selected. The panel is closed with **Close filters**. All functions described in this manual are available on a telephone.

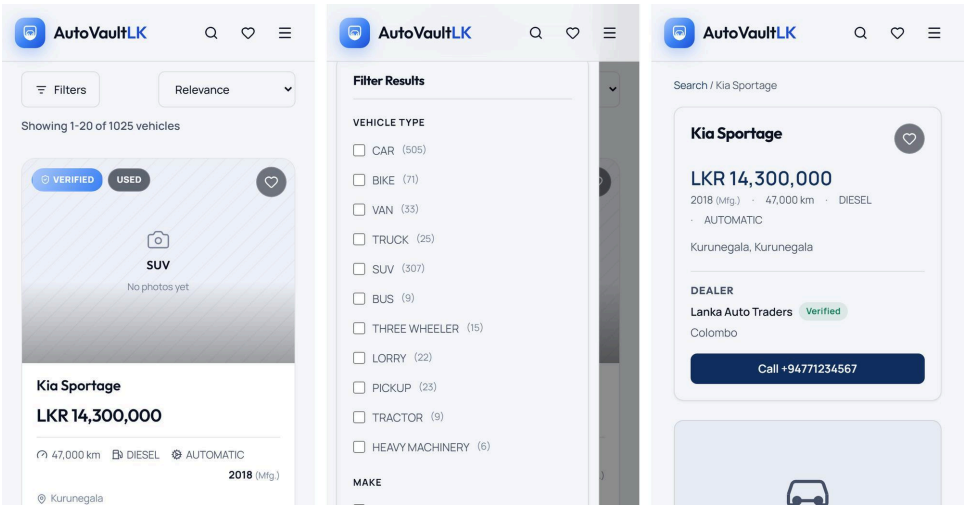


Fig. 4. Mobile layout: search results (left), filter panel (centre) and vehicle detail (right).

3. Searching for Vehicles

3.1 Search Page

Search is opened with **Browse**, with the search icon in the header or with the search panel on the home page. The search page, shown in Fig. 5, consists of a filter panel on the left, the result cards in the centre and a sort menu at the top right. Results are listed 20 to a page together with the total count, and **Previous**, **Next** and page numbers at the foot of the list move between pages. Each card shows the photograph, make and model, price in Sri Lankan rupees (LKR) with a **Negotiable** label where the dealer accepts offers, mileage, fuel type, transmission, year of manufacture and location. A **Verified** badge marks vehicles listed by dealers whose documents have been checked by an administrator, and a placeholder indicates that the dealer has not yet supplied a photograph.

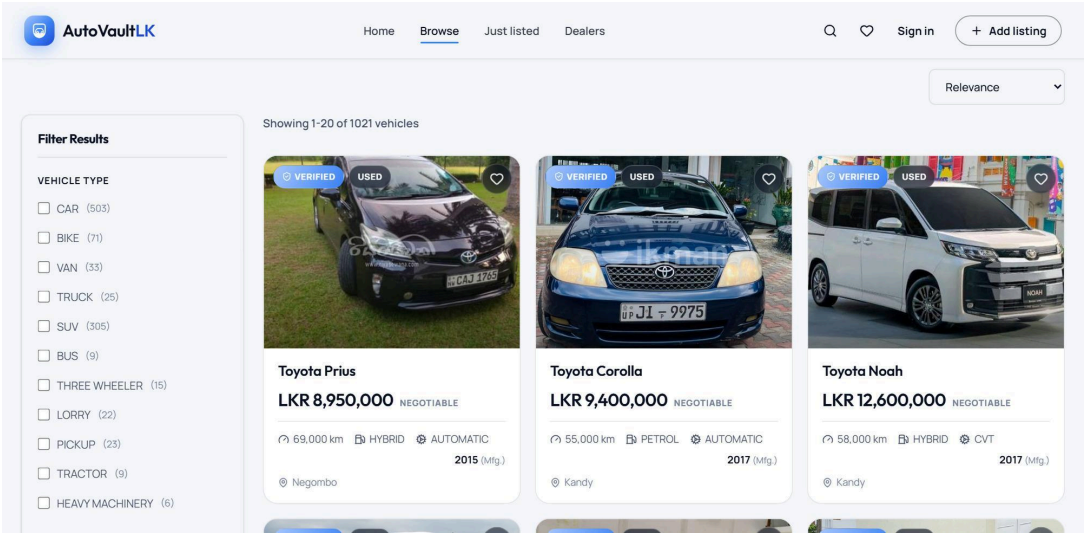


Fig. 5. Search page with the filter panel, sort menu and vehicle cards.

3.2 Natural-Language Search

No filter needs to be set to begin. A sentence such as “Toyata Corrola under 8.5m deisel” is interpreted as a make, model, price limit and fuel type despite the spelling errors, and the interpreted query appears as a chip above the results, as shown in Fig. 6. The chip can be removed to cancel the search. Words that cannot be interpreted are reported in a notice and ignored, and the results can then be refined with the filter panel.

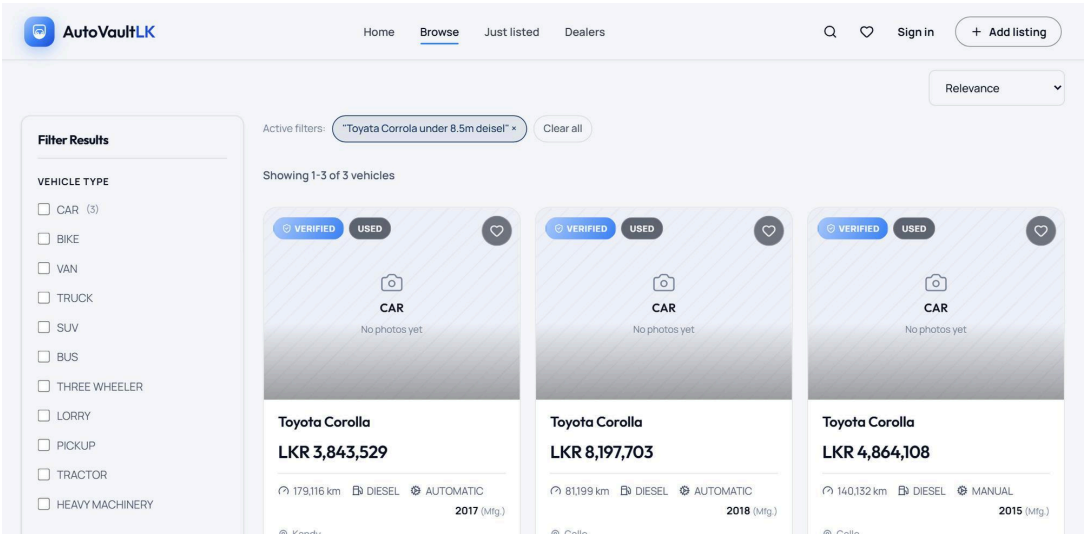


Fig. 6. Natural-language search with the interpreted query shown as a chip.

3.3 Filters

The filter panel narrows the results, and each change takes effect immediately. The number beside an option indicates how many vehicles match it. The principal filters are listed in Table 2. Filters that apply only to certain vehicles appear when they are relevant, so that seats and drive type are offered for passenger vehicles, engine class for motorcycles and load capacity for trucks and lorries. Every applied filter is also shown as a removable chip under **Active filters**, and **Clear all** restores the complete list.

Table 2. Principal search filters.

Filter	Options
Vehicle type	Car, bike, van, truck, SUV, bus, three-wheeler, lorry, pickup, tractor and heavy machinery.
Make and model	Models offered depend on the makes and vehicle types selected.
Price and year	Minimum and maximum values.
Condition	New, used or reconditioned.
Maximum mileage	Ceilings from 10,000 km to 150,000 km.
Fuel and transmission	Petrol, diesel, hybrid, electric or CNG; manual, automatic, CVT or semi-automatic.
District and body type	District of the vehicle; sedan, hatchback, SUV, wagon, coupe, convertible, pickup, minivan, scooter or motorbike.
Previous owners	First owner only, or up to two or three owners.
Yes-or-no options	Negotiable price only, verified dealers only, confirmed registration year only, and sunroof.

3.4 Sorting and Unsuccessful Searches

Results can be ordered by relevance, most recent, price in either direction, newest year or lowest mileage through the sort menu. The **Just listed** item of the main menu opens the search page in most-recent order. When a search finds nothing, the system relaxes it step by step, for example by dropping a mileage range or a keyword, and states what was relaxed, as shown in Fig. 7. If nothing matches even after relaxation, an explanatory message and a **Clear all filters** button are displayed.

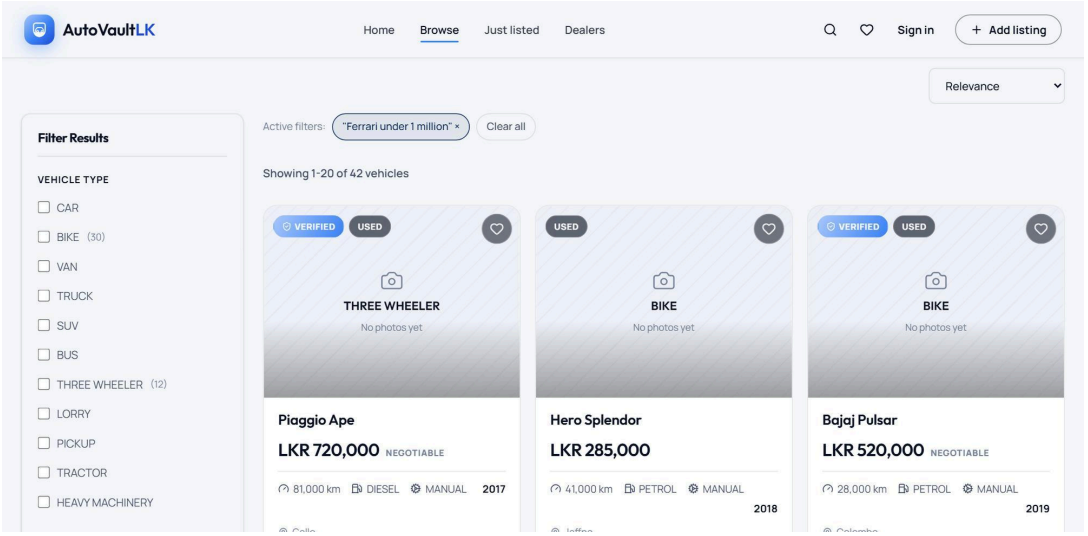


Fig. 7. Widened search that lists the closest vehicles when no exact match exists.

4. Viewing a Vehicle

4.1 Vehicle Detail Page

Selecting a card opens the detail page, shown in Fig. 8. The left side presents the photograph gallery, the description written by the dealer and a specification list that includes the vehicle type, year, mileage, condition, fuel, transmission, colour, engine capacity, number of previous owners and any attributes specific to the type of vehicle, such as seats or load capacity. The summary panel on the right states the price, the key facts and the location, and it identifies the dealer. A green **Verified** label confirms that the dealer has been checked, and a **Call** button appears when the dealer has given a telephone number. On a telephone the button starts the call directly.

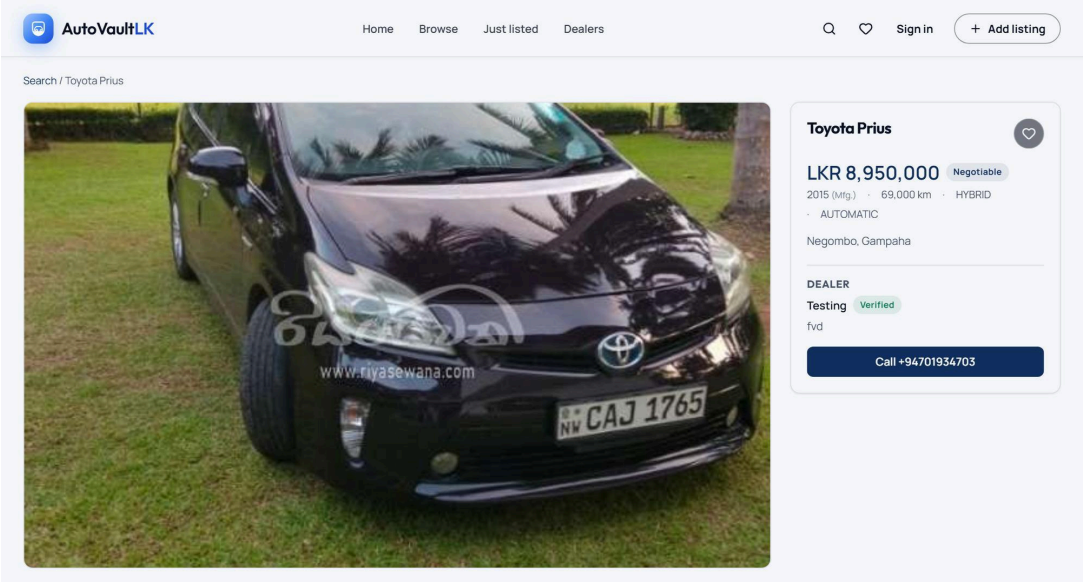


Fig. 8. Vehicle detail page.

4.2 Similar Vehicles and Unavailable Listings

A **Similar vehicles** strip at the foot of the page, shown in Fig. 9, suggests comparable listings. The strip is omitted when nothing similar exists. A vehicle that has been sold, archived or removed produces a page headed "Listing not available", which offers a link back to the search page.

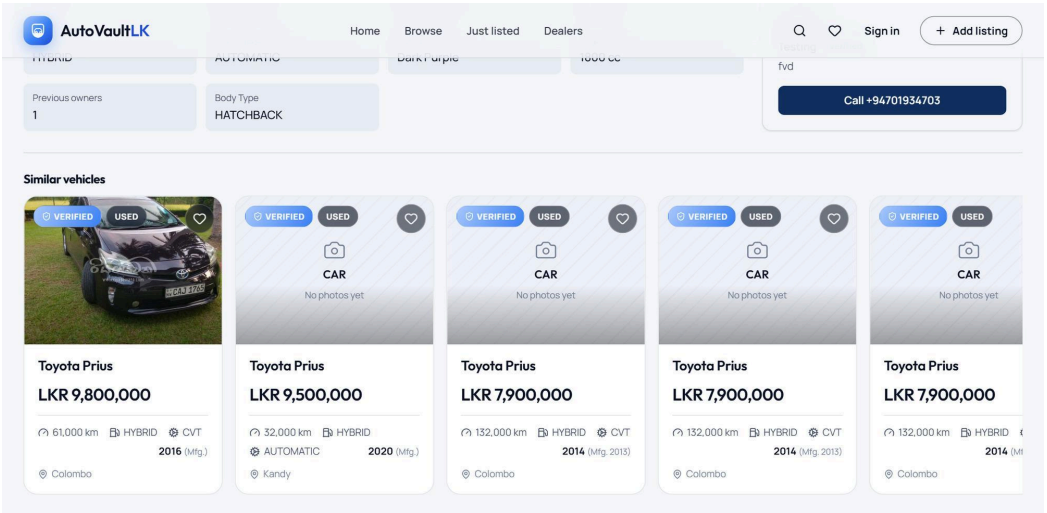


Fig. 9. Similar vehicles strip at the foot of the detail page.

4.3 Contacting the Dealer

AutoVaultLK connects buyers and dealers but does not take part in the sale. Viewing, negotiation and payment are arranged directly with the dealer. The vehicle and its registration and ownership documents should be inspected in person before any payment is made, and the **Verified** label should be treated as a check on the dealer rather than a guarantee of the condition of the vehicle.

5. Account Management

5.1 Registration

A buyer account is created from the **Create one** link on the sign-in page. The form, shown on the left of Fig. 10, requests a full name of at least two characters, an email address, and a password that is typed twice. The password must contain at least eight characters, which is the minimum length recommended in current guidance on authentication [4], and it must also include at least one lowercase letter, one uppercase letter and one digit. A field that is incomplete or invalid is marked with a message beneath it, as in Fig. 11, and the account is created only when every field is valid.

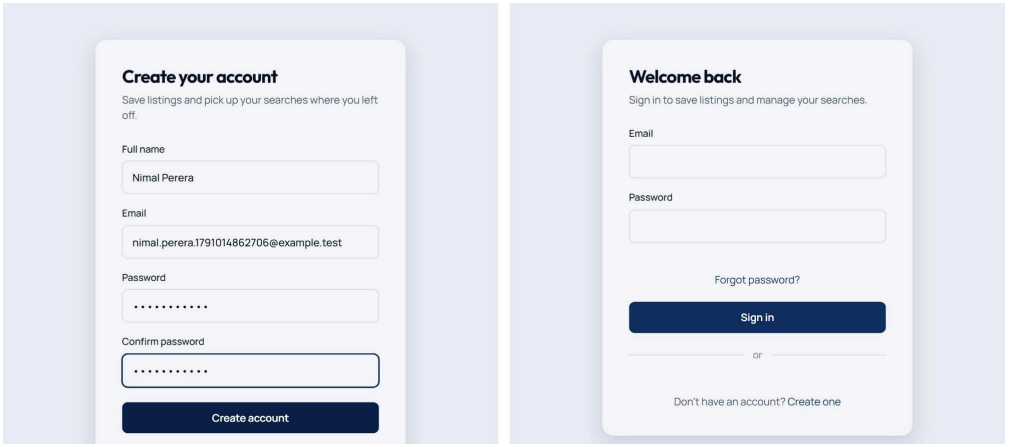


Fig. 10. Completed registration form (left) and sign-in page (right).

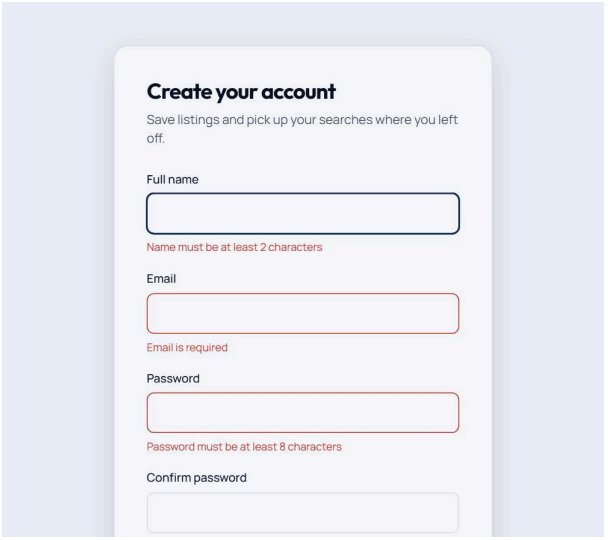


Fig. 11. Validation messages shown on an incomplete registration form.

5.2 Email Verification

After registration, the platform sends a verification email and displays the notice shown on the left of Fig. 12. The account becomes usable when the link in that email is opened, which produces the page shown on the right. If the message does not arrive, the spam folder should be checked and a new message requested with the resend button. Only the newest link is valid, and a link expires after a limited period, so an old message that fails should be replaced

by requesting another. An unverified account cannot sign in, and an attempt to do so displays a message together with a button for sending the verification email again.

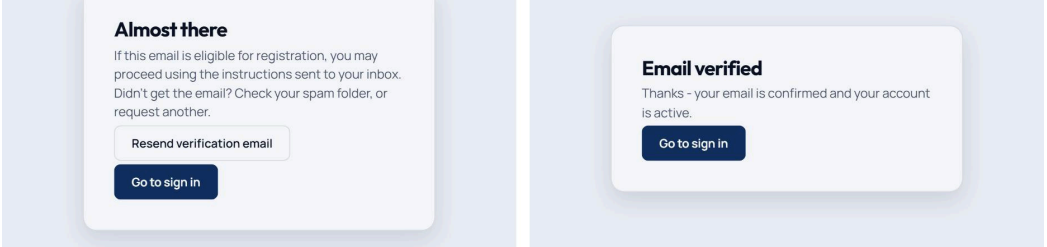


Fig. 12. Confirmation notice after registration (left) and verified email page (right).

5.3 Signing In and Password Recovery

A buyer signs in from the **Sign in** link with the registered email address and password and is returned to the page that was being visited, or to the search page. A wrong password produces the message shown on the left of Fig. 13. Repeated failures cause sign-in to be suspended temporarily, after which the attempt may be repeated or the password reset. A forgotten password is recovered through **Forgot password?**, which requests the email address and sends a reset link. The reply is identical whether or not an account exists, as on the right of Fig. 13, so that the page cannot be used to discover which addresses are registered. The reset link remains valid for about an hour, and only the newest link works.

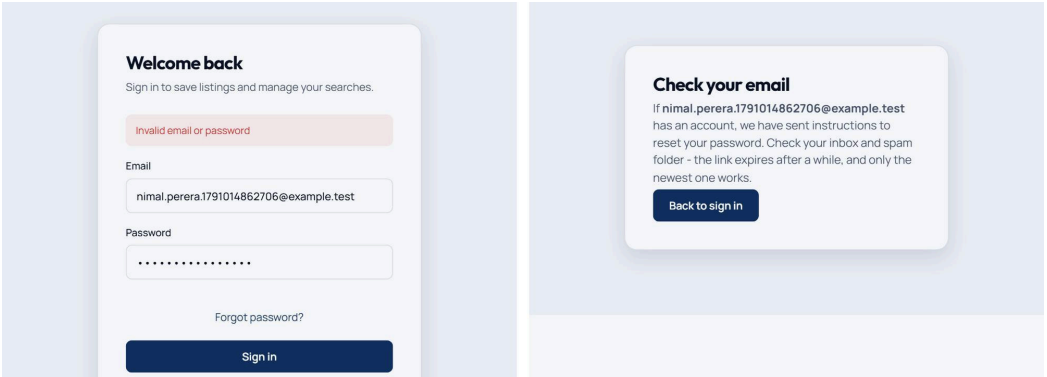


Fig. 13. Sign-in error message (left) and password reset confirmation (right).

5.4 Sessions and Signing Out

A session is renewed automatically while the platform is in use. A session that cannot be renewed, for example after a long absence, ends with a request to sign in again, and the page being visited is reopened afterwards. On a shared computer, **Sign out** in the account menu should always be selected on completion.

6. Saved Vehicles

6.1 Saving and Removing Vehicles

A signed-in buyer saves a vehicle by selecting the heart icon on a card or on the detail page, and a message at the top right confirms the action, as in Fig. 14. Selecting the filled heart removes the vehicle from the list. A guest who selects the icon is informed that signing in is required and is taken to the sign-in page.

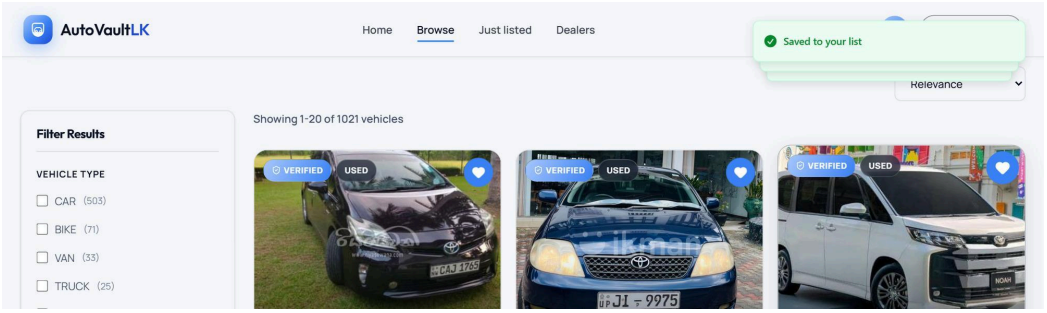


Fig. 14. Confirmation message displayed after a vehicle is saved.

6.2 Saved List

The saved list opens from the heart icon in the header or from **Saved vehicles** in the account menu, and it presents the saved vehicles as cards, as shown in Fig. 15. A card opens the detail page, and its heart removes the vehicle, which then disappears from the list at once. An empty list displays a short message and a **Browse vehicles** button.

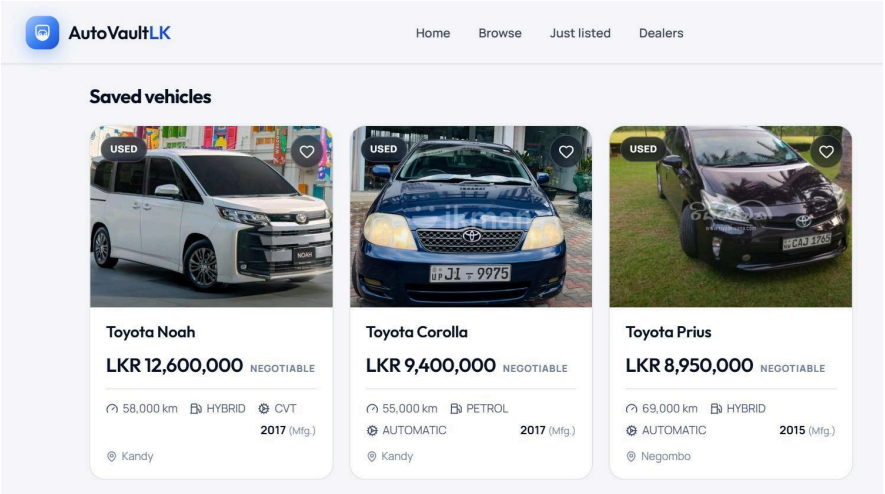


Fig. 15. Saved vehicles page.

7. Troubleshooting and Safe Use

7.1 Common Difficulties

The most frequent difficulties and their remedies are listed in Table 3.

Table 3. Common difficulties and remedies.

Difficulty	Remedy
Verification email not received	Check the spam folder, confirm that the address was typed correctly, and use the resend button.
Verification or reset link fails	The link has expired or has been replaced. Request a new one and use the newest message.
Sign-in suspended	Wait for a while, then try again or reset the password.
Search finds nothing	Remove a filter chip, widen the price range, or select **Clear all**.

Difficulty	Remedy
Part of the sentence is ignored	Use the filter panel for the details that were not understood.
A vehicle has disappeared	The vehicle has been sold or archived. Use the similar vehicles or the search page.
No Call button on a listing	The dealer has not given a number. Choose another vehicle or another listing of the same dealer.

7.2 Safe Use

Passwords should be unique to the platform and never shared. Emails that were not requested should be ignored, because the platform sends messages only to confirm an address or to reset a password that was requested. Buyers should not pay before inspecting the vehicle and its documents, and should be cautious of any request for money in advance.

References

[1] Systems and Software Engineering—Design and Development of Information for Users, ISO/IEC/IEEE Standard 26514:2022, 2022.

[2] Group 25, “Software Requirements Specification: Second-Hand Vehicle Marketplace with Intelligent Search and Automated Inventory Processing,” Dept. Comput. Sci. Eng., Univ. Moratuwa, Moratuwa, Sri Lanka, Project Rep. PID 11, Ver. 1.0, 2026.

[3] Group 25, “Software Architecture Document: Second-Hand Vehicle Marketplace with Intelligent Search and Automated Inventory Processing,” Dept. Comput. Sci. Eng., Univ. Moratuwa, Moratuwa, Sri Lanka, Project Rep. PID 11, Ver. 2.0, Aug. 10, 2026.

[4] P. A. Grassi et al., “Digital identity guidelines: Authentication and lifecycle management,” Nat. Inst. Standards Technol., Gaithersburg, MD, USA, Spec. Publ. 800-63B, Jun. 2017, doi: 10.6028/NIST.SP.800-63b.